


Thm 4.3

If g is continuous,

$\{T_n\}_{n=1}^{\infty}$ sequence of R.V's

$$T_n \xrightarrow{P} C$$

$$g(T_n) \xrightarrow{P} g(C)$$

$$h(x, z) = x - z^2$$

$$h(x) = \frac{x}{1-x}$$

This is continuous $[0, 1)$

Bound $\Pr(|\bar{X} - \mu| \leq C) \geq 0.9$

$$\begin{aligned} 1 - \Pr(|\bar{X} - \mu| \geq k\sigma) &\geq 1 - \frac{1}{k^2} \\ &= \Pr(|\bar{X} - \mu| \leq k\sigma) \geq 1 - \frac{1}{k^2} \end{aligned}$$

$$\text{std}(\bar{X}) = \sqrt{\frac{\sigma^2}{n}} = \frac{\sigma}{\sqrt{n}} = \frac{3}{10}$$

$$E(\bar{X}) = \mu$$

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

Restatement of Chebyshev's II inequality

Let X be a R.V. with $E(X)$ and $\text{Var}(X)$ finite.
Then for any real number $k > 0$

$$\Pr(|X - E(X)| \geq k \sqrt{\text{Var}(X)}) \leq \frac{1}{k^2}$$